

Total No. of Questions : 8]

SEAT No. :

PB2541

[Total No. of Pages : 2

[6263] 440

B.E. (Honours) (Artificial Intelligence and Machine)

SOFT COMPUTING AND DEEP LEARNING

(2019 Pattern) (Semester - VIII) (410303)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6 and Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.

- Q1)** a) Elaborate Roulette Wheel's Selection algorithm in Genetic Algorithm. [6]
b) Describe Genetic Algorithm Requirements [6]
c) What is Mutation in Genetic Algorithm? Explain with examples. [6]

OR

- Q2)** a) Distinguish in between Symbolic AI and Genetic Algorithms. [6]
b) Write note on Evolving Neural Networks with example. [6]
c) List the applications of Genetic Algorithms with explanation of one use case. [6]

- Q3)** a) Compare Biological Neural Network (BNN) with Artificial Neural Network (ANN). [5]
b) Describe the architecture of artificial neural network (ANN). [6]
c) Explain the process of Modeling a Neuron. [6]

OR

- Q4)** a) Draw and explain General Architecture of McCulloch-Pitts Neuron. [6]
b) Write detailed note on Multi-Layer Perceptron. [5]
c) Compare Perceptron vs. Gradient Descent Rule. [6]

P.T.O.

- Q5)** a) Explain the working principle of Convolution Neural Network (CNN) with suitable diagram. [6]
- b) Describe the Long-Short Term Memory (LSTM) algorithm with appropriate diagram. [6]
- c) Write detailed note on Recurrent Neural Network (RNN). [6]

OR

- Q6)** a) Explain the role of classification in deep learning. [6]
- b) Elaborate the use case of Deep Learning. [6]
- c) Describe the working principle of Binary Classifier in Neural Networks. [6]

- Q7)** a) Write note on Vanishing Gradient in Deep Learning. [6]
- b) Explain ReLU (Rectified Linear Unit) Activation Function. [5]
- c) Describe Saturating Gradient Problem with example. [6]

OR

- Q8)** a) Write note on importance of the Pooling Layer in CNN. [5]
- b) List the applications of Deep Learning and explain any one application in details. [6]
- c) Explain the working principle of Deep Learning. [6]

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